Class 5

In the second price auction,

$$u_i(v_i, b_{-i}) \geq u_i(b_i, b_{-i})$$

$$\forall\, b_{-i} \in [0, \infty)^{n-1}$$

$$\forall\, b_i \in [0, \infty)$$

Three players, for player 1, $v_1 = 11$

$$b_2 = 10, \quad b_3 = 8$$

The above theorem does not hold in first price auction.

Assume $n \geq 3$ and check if the above theorem holds for the third price auction.

Max-min value and max-min strategies

A player plays strategy s_i

$$\min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$$

For player i, by playing his own, he/she can ensure at least $\max_{s_i \in S_i} \min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$

This is called the max-min value for player i.

And, any strategy $s_i \in \arg\max_{s_i \in S_i} \left(\min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) \right)$

is called his/her max-min strategy

If a player plays a max-min strategy, he/she will get at least his max-min utility.

If the strategy spaces are infinite, then max-min value is $\sup_{s_i \in S_i} \inf_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$

- $$u_i(x_1, x_2, x_3) = \begin{cases} 0 & \text{if } |x_i - T(x_1, x_2, x_3)| > |x_j - T(x_1, x_2, x_3)| \text{ for } j \\ \dfrac{1}{|\{k \mid k \in \arg\min |x_k - T(x_1, x_2, x_3)|\}|} & \text{O.w.} \end{cases}$$

$x_1 = 10$, $\min_{(x_2, x_3) \in [0,100]^2} u_i(10, x_2, x_3) = 0$

$x_1 = 0$ $\min_{(x_2, x_3) \in [0,100]^2} u_i(0, x_2, x_3) = 0$

max min value is 0.

Any strategy is a max-min strategy.

- for second price auction.

max-min value is 0 and bidding your valuation is your max-min strategy.

Playing anything less than your valuation is a max-min

strategy.

- Suppose player i's max-min value is v_i

$G = \langle N, (S_k)_{k \in N}, (u_k)_{k \in N} \rangle$ $\quad G' = \langle N, (S_k')_{k \in N}, (u_k)_{k \in N} \rangle$

where $S_k' = S_k$ for all $k \in N \setminus i$

$S_i' = S_i \setminus \{t_i\}$

where t_i is dominated by some strategy in G.

max-min value of player i in G is v_i

max-min value of player i in G' is also v_i

$S_k' = S_k$ for all $k \in N \setminus j$

$S_j' = S_j \setminus \{t_j\}$ where t_j is dominated by some strategy.

In this case, the max-min value of player i will increase.

Two-player Zero-sum game

A game $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ is called a two-player zero sum game if

- $N = \{1, 2\}$
- $u_1(s_1, s_2) + u_2(s_1, s_2) = 0 \quad \forall (s_1, s_2) \in S_1 \times S_2$

1 \ 2	P	Q
U	$u_1(U,P)$	$u_1(U,Q)$
D	$u_1(D,P)$	$u_1(D,Q)$

